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ARCHITECTURE OVERVIEW · EMBEDDEDLEND

The Underwriting Bridge, in One Picture

A companion to the PRD — the consent-based SME underwriting architecture rendered as one clear system diagram: data in, decision made, loan out, repayment loop.

EmbeddedLend

Rwanda → East African Community

PRD companion · Architecture visualization

v1.0 · August 2026

prd-embeddedlend.md (full architecture & MVP slice)

The Architecture at a Glance

The full architecture of the PRD, rendered as one clear flow. The lender (top right) requests a decision; the borrower (top left) consents. Consented data flows from Open Finance and mobile-money rails into a deterministic Decision Engine, whose AI layer explains the call back to the lender. An approved loan is disbursed over eKash and repayment history loops back to sharpen future scores, all under compliance.

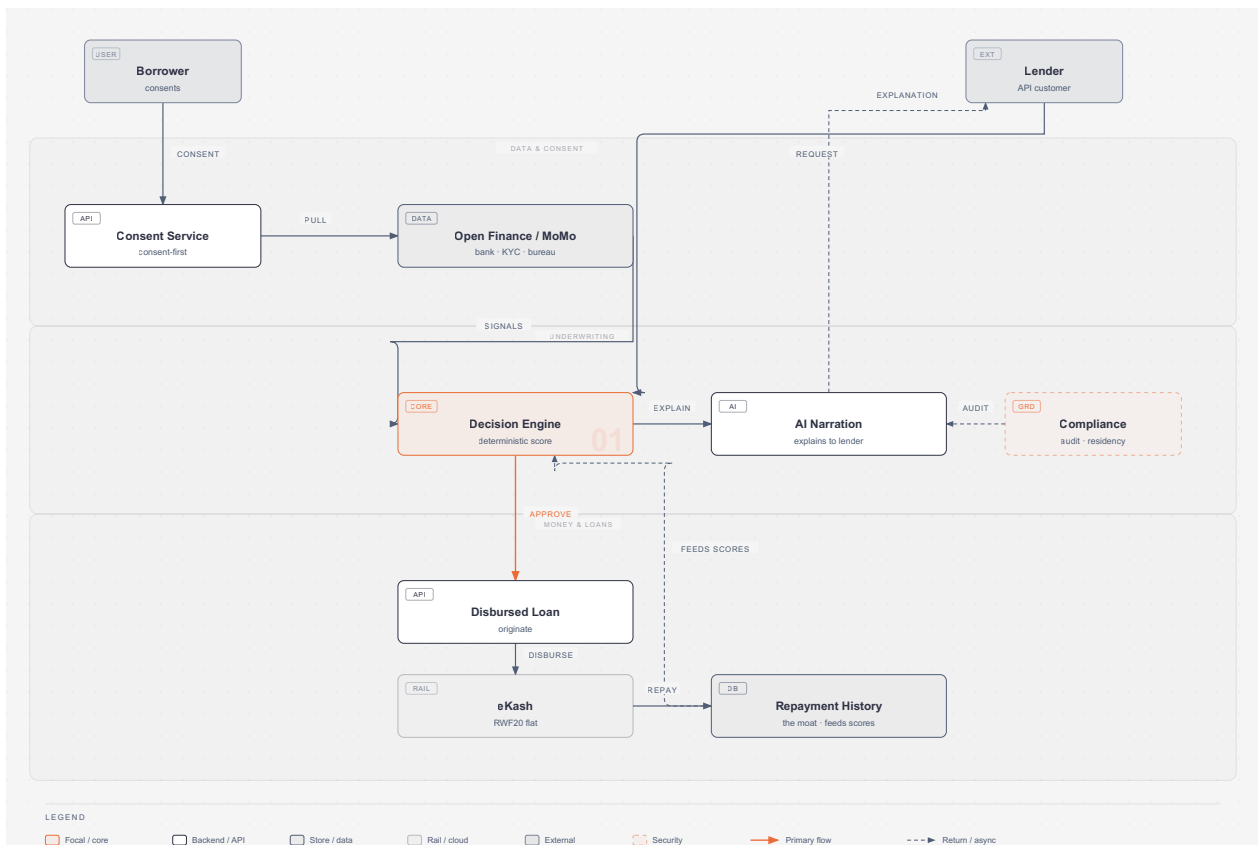


Figure 1 — EmbeddedLend, the system at a glance (editorial overview)

Interactive companion: open <architecture/diagram.html> in a browser for the animated, explorable version. The **.excalidraw** file is editable. The **.svg** is the scalable static source.

Reading the Diagram

1. What flows in (left side)

- **Borrower Consent** — the legal gate: data is pulled only with the borrower's opt-in (Law 058/2021 + GDPR-style rights).
- **Open Finance** — BNR's data-sharing rail (voluntary Phase 1 now, mandatory Phase 2 2026–28).
- **MoMo / Bank APIs** — MTN MoMo, Airtel, bank records, MTN KYC.
- **eKash Rail** — the national payment switch (RWF20 flat, RWF10m cap, one integration).

2. The underwriting core (center)

- **Data & Consent** — consented pulls, aggregated ledger.
- **Signals** — deterministic cash-flow features: income stability, concentration, seasonality, repayment capacity.
- **Decision** — a deterministic, explainable score plus a full decision trace. *Numbers computed by code; narrative by AI. No black-box scores.*
- **Loans** — origination and repayment scheduling.

3. The gate and the loop

- **Approve?** — policy rules (score + limits). "Declined / missing data" routes back for retry.
- **Disbursed Loan** — the output, sent over eKash.
- **Repayment loop** — repayment history feeds back into future scores. This is the moat: the aggregated credit-history dataset no single lender owns.

4. The side panels

- **Lender Personas (who pays)** — NDFSP (full-stack underwriting), MFI/SACCO (scoring + decision APIs), Vertical SaaS (embedded credit flow).
- **Compliance** — consent records + revocation, tamper-proof audit, data residency — the regulator's comfort layer.
- **Money Movement** — disburse and repay over eKash: one integration, cheap, instant, interoperable.

The MVP Slice (flagged in the PRD)

The MVP slice lives in the PRD (Section 8), not here: one consent flow, one data source (mobile-money cash-flow), the scoring engine, the decision service, the eKash connector, and the core endpoints (consent, data/summary, score, decisions, loans, disbursements,

repayments). This companion stays diagram-forward — the whole system in one picture; the PRD holds the build order.

Companion to prd-embeddedlend.md (full architecture & MVP slice). Diagram generated in hand-drawn style; the walkthrough mirrors the PRD's architecture, data flow, personas, and compliance at diagram level. Build order and endpoints live in the PRD, not here. Open question carried forward: lender willingness-to-pay, to be validated in the Field stage.